

ITT-036: Programming Assignment 1

Title: Digital Signal Generator and Encoder Visualizer

Students :

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1. Language and Libraries Used

- **Programming Language:** C++
- **Graphics/Visualization:** OpenGL (for 2D graphics plotting) and FreeGLUT (OpenGL Utility Toolkit wrapper for C++ cross-platform windowing and event management)
- **C++ Standard Library:** <iostream>, <vector>, <string>, <cmath>, <algorithm>
- **OpenGL / Graphics:** <GL/freeglut.h>

2. How to Run

Ensure `freeglut.dll` is present in your run folder on Windows :

1. **Compile:** `G++ code.cpp -o Signal_Generator.exe -Iinclude -Llib/x64 -lfreeglut -lopengl32 -lglu32`
2. **Run:** `.\Signal_Generator.exe`

3. Usage

- Enter digital/analog input as prompted.
- Choose encoding and (if analog) modulation.
- Output, analysis and animation shown in terminal and OpenGL window.

4. Assumptions

- All digital inputs are valid bitstrings (0/1 only).
- All analog sample values are valid real numbers.
- PCM uses 16 uniform levels (4 bits/sample).
- Input length is manageable for in-memory execution.

5. References

- Dr Iqra Altaf Gilani Class Notes - Module 5: Digital Transmission
- Forouzan, B. A. "Data Communications and Networking", 4th/5th Edition.
- FreeGLUT Documentation/Community Examples.